

Analysis Report

Approach to data extraction and analysis:

1. Since we are working with images of bill receipts and revenue records—without relying on LLMs or vLLMs—accurate structure identification is essential, especially for tabular data.
2. To begin with, extracting text from images requires a highly accurate OCR engine to preserve the structure effectively. I evaluated Tesseract OCR, EasyOCR, and PaddleOCR, and found that **PaddleOCR consistently produced the most accurate results**.
3. For table layout extraction, I used an open-source Python library called **img2table**. While I had previously tested the layout extraction capabilities of both Tesseract and PaddleOCR, neither of them could reliably detect and preserve table structures, making it difficult to convert them accurately into Excel or CSV formats.
4. After further research and testing of various OCR-based table extraction tools, **img2table** provided the best performance. Additionally, I integrated PaddleOCR with **img2table** to enhance the accuracy of parsing tables from images into structured Excel files.

Enrichment:

1. Feedback and comments needed to be enriched with individual-level analysis to derive meaningful insights. To achieve this, I developed analyzers to extract features such as sentiment scores, sentiment polarity, descriptive adjectives, and more.
2. For images, enrichment was achieved through structured data extraction and parsing, enabling meaningful interpretation of the visual content.

Analysis:

I placed greater emphasis on generating the analysis report. Recommendations were included specifically for the tabular revenue records. Since we are not using LLMs, these recommendations are partially hardcoded. However, this approach can be easily extended to other types of data as well, allowing for flexible integration of additional rule-based recommendations.

I also integrated Matplotlib to generate charts and histograms as part of the analysis. The resulting images are stored in Supabase Object Store, and the corresponding image URLs are returned along with the response. This approach supports easy extensibility of the analysis output.

Additionally, I avoided tying the system to a specific data format to ensure flexibility and generalization. For instance, when extracting key-value pairs from bills, the analysis focuses only on common, meaningful keywords such as **total**, **guests**, etc., allowing the solution to adapt to varying bill formats without requiring structural changes.

Analysis examples include:

[Tabular Revenue records](#), [Bill receipt](#), [Feedbacks](#), [comments](#)

Appendix

1. Tabular Revenue records:
Screenshot:

Bean & Brew Coffee Shop - Monthly Revenue Record: April 2025			
Date	Day of Week	Total Revenue	Transaction Count
2025-04-01	Tuesday	\$1,247.65	143
2025-04-02	Wednesday	\$1,312.80	151
2025-04-03	Thursday	\$1,389.25	159
2025-04-04	Friday	\$1,456.70	172
2025-04-05	Saturday	\$2,103.45	206
2025-04-06	Sunday	\$1,875.30	187
2025-04-07	Monday	\$1,195.40	138
2025-04-08	Tuesday	\$1,256.85	145
2025-04-09	Wednesday	\$1,303.55	152
2025-04-10	Thursday	\$1,378.90	160
2025-04-11	Friday	\$1,467.25	173
2025-04-12	Saturday	\$2,235.10	211
2025-04-13	Sunday	\$1,923.75	190
2025-04-14	Monday	\$1,178.90	135
2025-04-15	Tuesday	\$952.35	110
2025-04-16	Wednesday	\$1,289.70	148
2025-04-17	Thursday	\$1,392.55	161
2025-04-18	Friday	\$1,352.15	157
2025-04-19	Saturday	\$2,056.80	201
2025-04-20	Sunday	\$1,835.95	183
2025-04-21	Monday	\$1,267.80	146
2025-04-22	Tuesday	\$1,351.05	155
2025-04-23	Wednesday	\$1,307.25	151
2025-04-24	Thursday	\$1,395.60	161
2025-04-25	Friday	\$1,487.30	175
2025-04-26	Saturday	\$2,312.40	218
2025-04-27	Sunday	\$1,941.85	191
2025-04-28	Monday	\$1,215.60	140
2025-04-29	Tuesday	\$1,265.90	146
2025-04-30	Wednesday	\$1,542.35	176

Request:

POST

http://localhost:8000/tabular_record

Send

Params

Auth

Headers (8)

Body

Scripts

Settings

Cookies

form-data

	Key		Value	Description	...	Bulk Edit
☒	file	File	Screenshot 2025-05...			
	Key	Text	Value	Description		

Response:

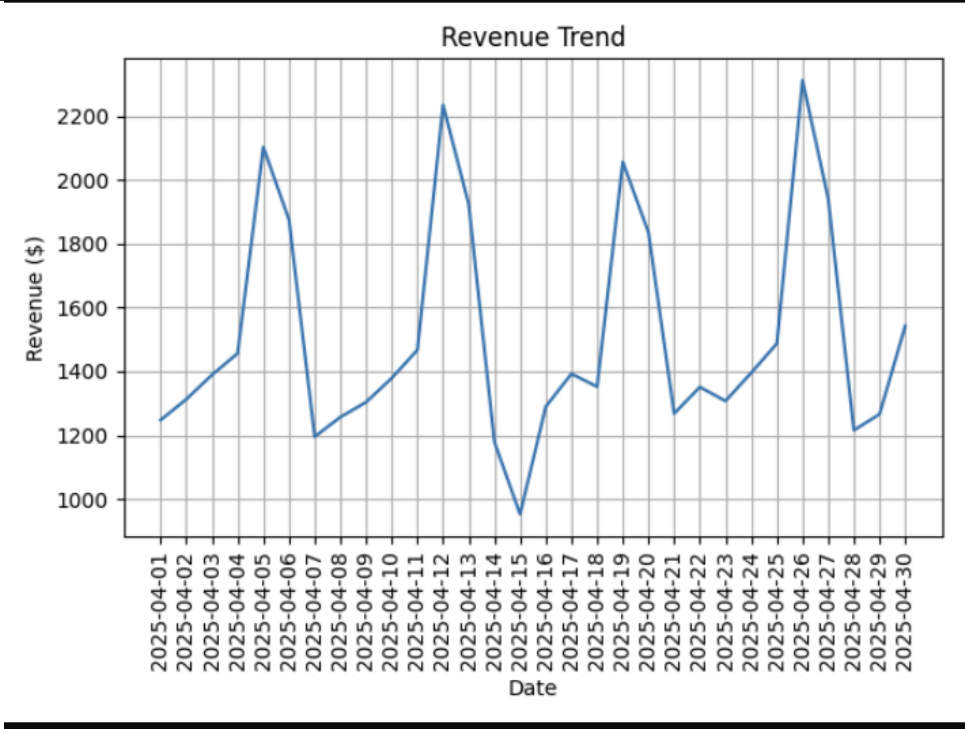
(structured and parsed data is also returned with the request but showing only the analysis part of the response)

Showing only the analysis and recommendation part here:

```
"excel_public_url":  
"https://wjidwijftszidvtxiqvv.supabase.co/storage/v1/object/public/test/76efce868d364ebfb8dfcbbfc478f17c_table-record.xlsx?"  
}  
"analysis": {  
  "trends": {  
    "daily_revenue_trend_image":  
"https://wjidwijftszidvtxiqvv.supabase.co/storage/v1/object/public/test/daily_trend_6d7a0792-2a55-4fe2-9bbe-2a9ffdf55d46.png?",  
    "weekly_average_revenue": {  
      "Friday": 1440.85,  
      "Monday": 1214.425,  
      "Saturday": 2176.9375,  
      "Sunday": 1894.2125,  
      "Thursday": 1389.075,  
      "Tuesday": 1214.76,  
      "Wednesday": 1351.1299999999999  
    },  
    "top_revenue_days": [  
      {  
        "Date": "2025-04-26",  
        "Total Revenue": 2312.4  
      },  
      {  
        "Date": "2025-04-12",  
        "Total Revenue": 2235.1  
      },  
      {  
        "Date": "2025-04-05",  
        "Total Revenue": 2103.45  
      }  
    ],  
  },  
  "engagement_metrics": {  
    "average_spend_per_transaction": 9.077839672946604  
  }  
},  
"recommendations": [  
  "Consider offering discounts or promotions on Mondays to increase traffic.",  
]
```

```
    "Optimize staffing on peak days like 2025-04-26 due  
to high revenue."  
]
```

Image_url:
https://wjidwijftsztidvtxiqvv.supabase.co/storage/v1/object/public/test/daily_trend_6d7a0792-2a55-4fe2-9bbe-2a9ffdf55d46.png?



2. Bill receipt

Screenshots:

Bill #2

BEAN & BREW COFFEE SHOP
123 Main Street, San Francisco
Tel: (415) 555-0123

RECEIPT #: 6734
DATE: 30/04/2025
TIME: 11:42 AM
SERVER: Sarah
TABLE: 7
GUESTS: 4

2 x Latte	\$11.00
1 x Cold Brew	\$5.75
1 x Chai Tea	\$4.50
2 x Chicken Sandwich	\$25.90
1 x Veggie Wrap	\$11.50
1 x Caesar Salad	\$12.95
2 x Chocolate Muffin	\$7.90

SUBTOTAL:

\$79.50

TAX (8.5%):

\$6.76

SERVICE CHARGE (18%):

\$14.31

TOTAL:

\$100.57

PAYMENT METHOD: Cash

AMOUNT TENDERED: \$120.00

CHANGE: \$19.43

Thank you for visiting!

COME AGAIN SOON

BEAN & BREW COFFEE SHOP
123 Main Street, San Francisco
Tel: (415) 555-0123

RECEIPT #: 6721
DATE: 30/04/2025
TIME: 08:37 AM
SERVER: Miguel
TABLE: 4
GUESTS: 2

1 x Cappuccino	\$5.50
1 x Americano	\$4.25
1 x Avocado Toast	\$9.95
1 x Granola Bowl	\$8.75

SUBTOTAL:

\$28.45

TAX (8.5%):

\$2.42

SERVICE CHARGE (18%):

\$5.12

TOTAL:

\$35.99

PAYMENT METHOD: Credit Card
CARD TYPE: VISA ****1234

TIP: \$7.00

TOTAL PAID: \$42.99

Thank you for visiting!

COME AGAIN SOON

Request:

POST ⌵ **Send** ⌵

Params Auth Headers (8) **Body** • Scripts Settings Cookies

form-data ⌵

	Key		Value	Description	⋮ Bulk Edit
☒	files	File ⌵	⚠ 2 files 📁		
	Key	Text ⌵	Value	Description	

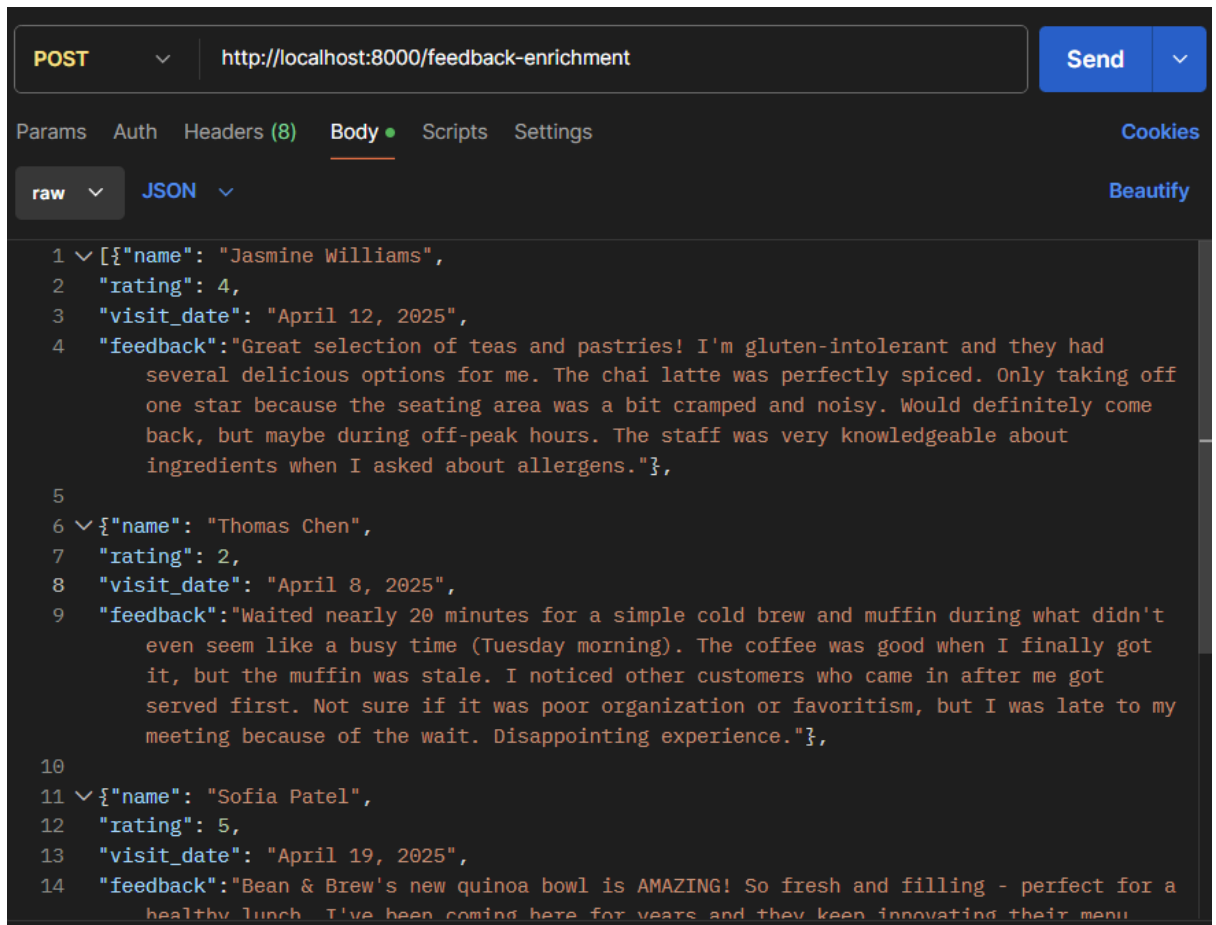
Response:

Only analysis part (structured and parsed data is also returned with the request but showing only the analysis part of the response):

```
"analysis": {  
  
  "bill_metrics": {  
  
    "total_bills": 2,  
  
    "total_revenue": 136.56,  
  
    "total_guests": 6,  
  
    "average_bill_amount": 68.28  
  
  },  
  
  "engagement_metrics": {  
  
    "average_guests_per_bill": 3.0,  
  
    "average_spend_per_guest": 21.56875  
  
  }  
  
}
```

3. Feedbacks

Request:



Response:(Below part is in text not an image)

(Showing only the analysis part, but the response also contains the sentiment score, polarity, etc for every feedback).

```
"analysis": {
  "total_feedbacks": 4,
  "average_rating": 3.25,
  "average_sentiment": 0.1993022760068215,
  "average_word_count": 69.0,
  "rating_distribution": {
    "1": 0,
```

```
        "2": 2,

        "3": 0,

        "4": 1,

        "5": 1

    },

    "sentiment_distribution": {

        "positive": 2,

        "neutral": 1,

        "negative": 1

    },

    "top_adjectives": {

        "good": 2,

        "gluten-intolerant": 1,

        "several": 1,

        "delicious": 1,

        "seating": 1

    },

    "chart_urls": [

        "https://wjidwijftszidvtxiqvv.supabase.co/storage/v1/object/public/test/daily_trend_bbdb25cb-647e-406e-977b-13c00f146d8d.png?",

        "https://wjidwijftszidvtxiqvv.supabase.co/storage/v1/object/public/test/daily_trend_affcec7f-80cc-47bb-a615-ebd5f46f61a3.png?"

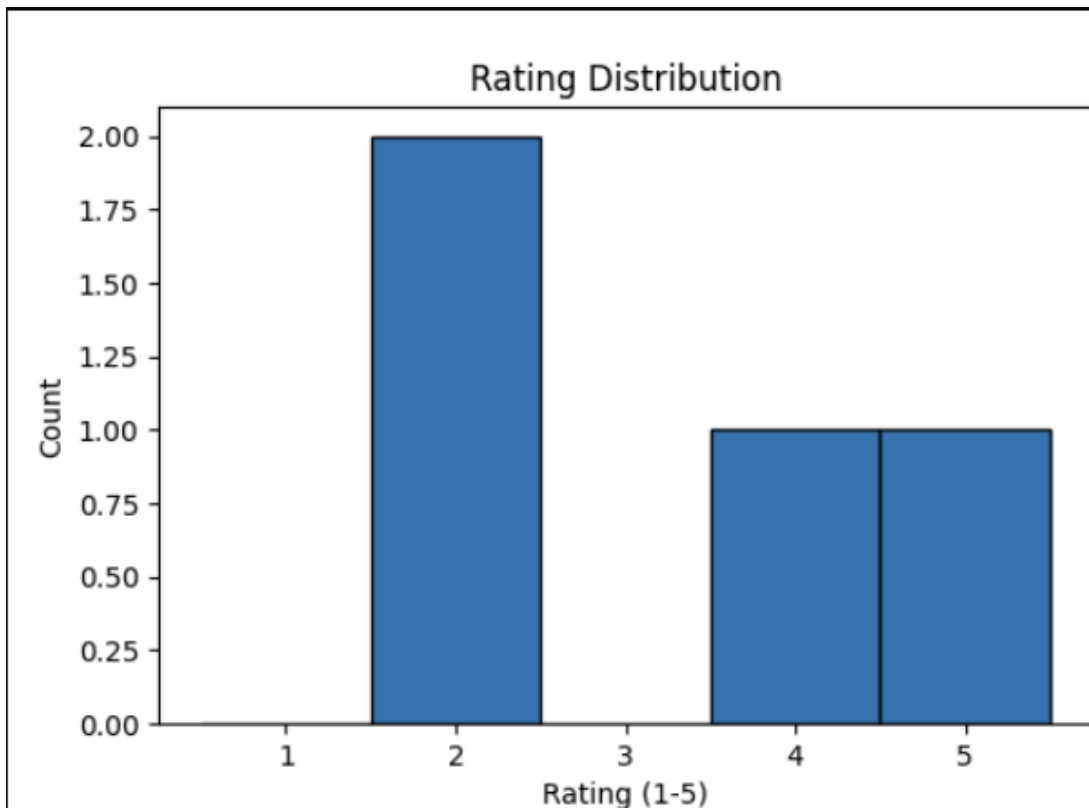
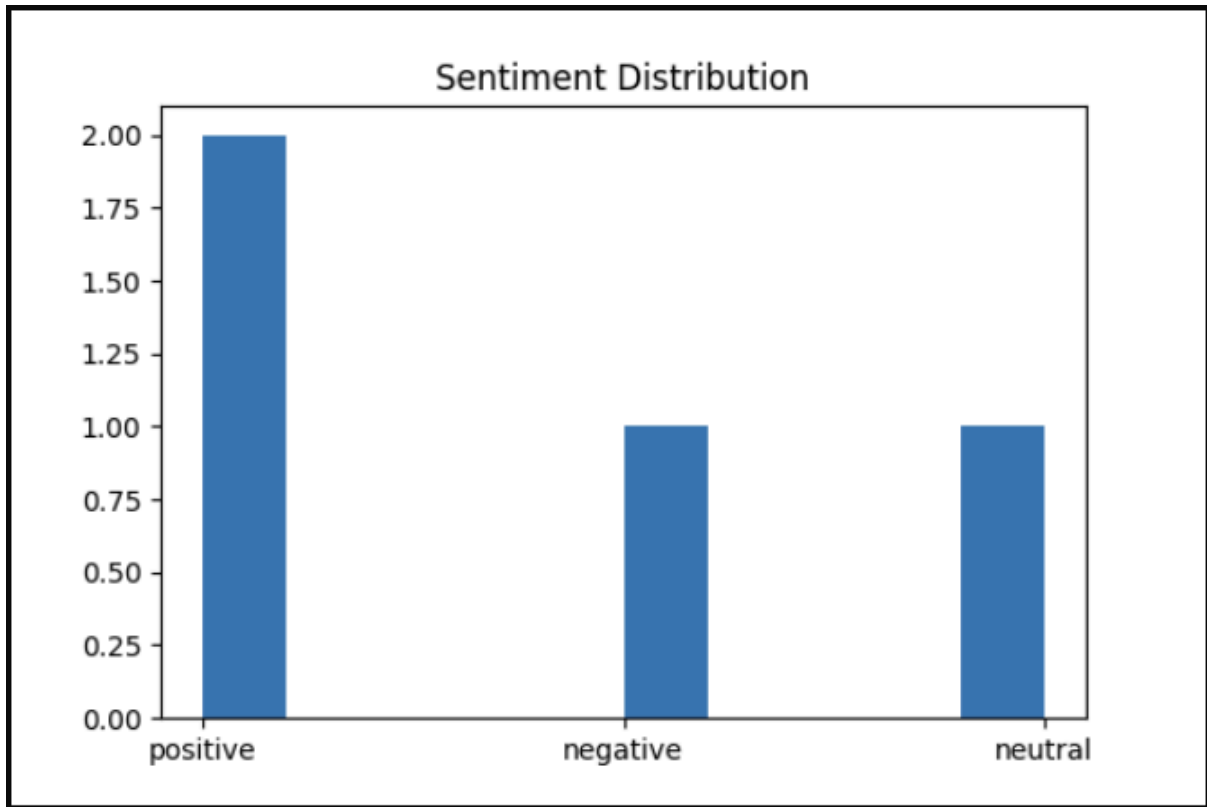
    ]

}
```

Image Urls:

1. https://wjidwijftszidvtxiqvv.supabase.co/storage/v1/object/public/test/daily_trend_bbdb25cb-647e-406e-977b-13c00f146d8d.png
2. https://wjidwijftszidvtxiqvv.supabase.co/storage/v1/object/public/test/daily_trend_affcec7f-80cc-47bb-a615-ebd5f46f61a3.png

3.



Comments-Analysis: Request:

GET

http://localhost:8000/comment-persist-analyze

Send

Params

Auth

Headers (6)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Response:

Full response(text):(Below part is in text not an image)

```
{
  "message": "Comment Feedbacks processed successfully",
  "data": [
    {
      "feedback": "Great experience",
      "sentiment": 0.8,
      "sentiment_label": "positive",
      "word_count": 2,
      "adjectives": [
        "Great"
      ]
    },
    {
      "feedback": "Nice Product",
      "sentiment": 0.6,
      "sentiment_label": "positive",
      "word_count": 2,
      "adjectives": []
    }
  ],
  "analysis": {
    "total_feedbacks": 2,
    "average_sentiment": 0.7,
    "average_word_count": 2.0,
    "sentiment_distribution": {
      "positive": 2,
      "neutral": 0,
      "negative": 0
    },
    "top_adjectives": {
      "Great": 1
    }
  }
}
```

```
    },  
    "chart_urls": [  
  
      "https://wjidwijftszidvtxiqv.supabase.co/storage/v1/object/public/test/daily_trend_2d9dec6a-147a-4318-b824-f6827f7260f5.png?"  
    ]  
  }  
}
```